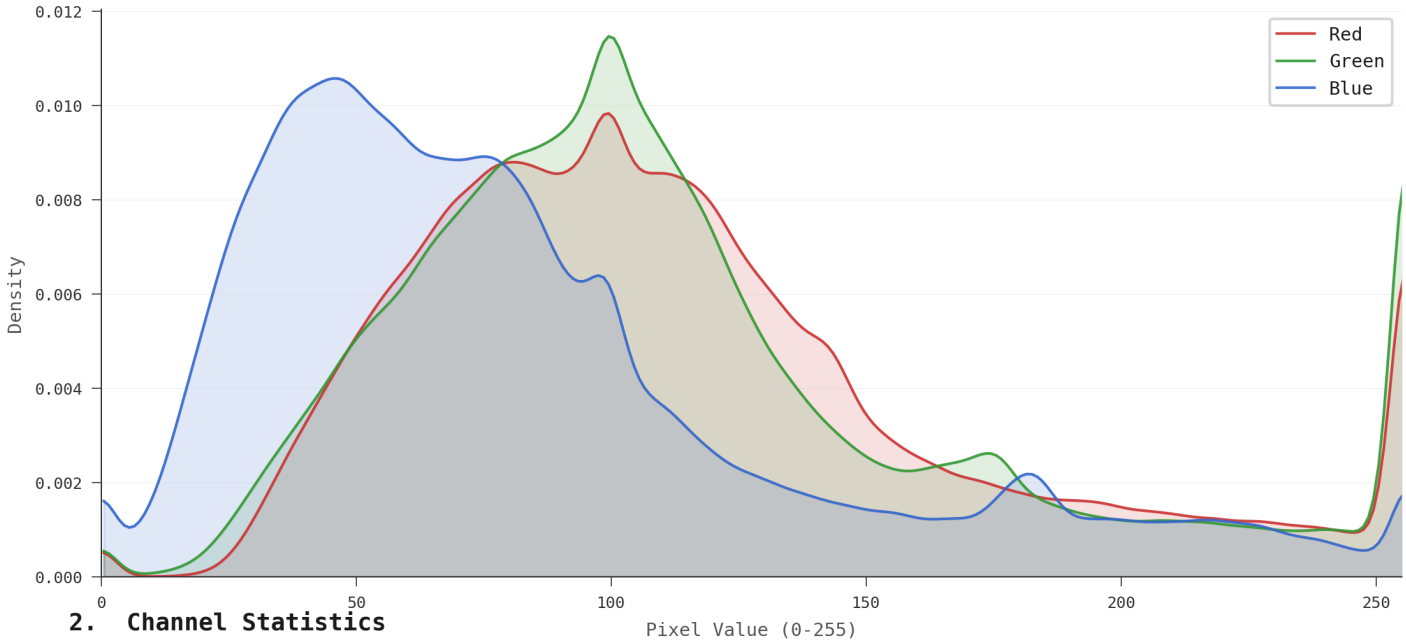
Reference Data Sheet: KODIM13

Subject: Mountain Stream and Rocks

768x512 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	112.2371	52.0095	2704.9840	0.5293	0.9326	0	255
Green	110.5166	53.5862	2871.4856	0.6601	0.9991	0	255
Blue	82.2896	54.7389	2996.3458	0.9736	1.2191	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.9818	0.9469
G	0.9818	1.0000	0.9639
B	0.9469	0.9639	1.0000

Pairwise Pearson Correlation Coefficients

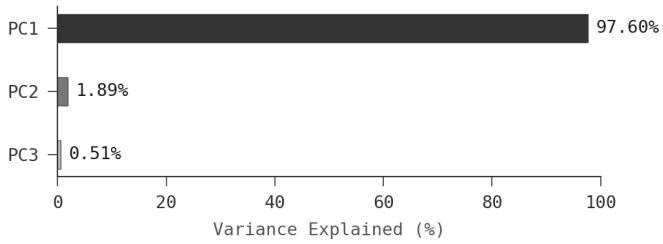
r(R-G)	=	**0.981787**
r(R-B)	=	**0.946925**
r(G-B)	=	**0.963893**
Avg |r|	=	**0.964202**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{2704.9909} & 2736.2400 & 2695.8463 \\
2736.2400 & \mathbf{2871.4929} & 2827.3473 \\
2695.8463 & 2827.3473 & \mathbf{2996.3535}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM13

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5614	-0.5821	**-0.5882**	8367.1677
PC2	-0.5688	-0.2449	0.7852	161.6290
PC3	-0.6011	0.7754	-0.1936	44.0405

6. Derived Metrics

Condition Number

189.99

(l1/l3) PC1/PC2: 51.77 | PC2/PC3: 3.67

Blue Channel Independence

3.4%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Blue

Loading: -0.5882 | Balanced

Classification: Near-Degenerate (PC1 = 97.60%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.894298	0.837180	0.865739
Green	0.897359	0.841415	0.869387
Blue	0.903094	0.852387	0.877740

8. Average Local Variance (3x3)

Channel	Value
Red	497.5990
Green	519.2852
Blue	492.1569

9. Redundancy Profile

Classification: Near-Degenerate | Condition Number: 189.99 | PC1: 97.60%

Correlation: R–G pair dominates at r = 0.981787 | G–B: 0.963893 | R–B: 0.946925

Blue Channel: 3.4% independent | 96.6% captured by PC1

Spatial: Avg autocorrelation: 0.870955 | Avg local variance: 503.0137

Eigenvector: Balanced | R (-0.5614), G (-0.5821), B (-0.5882) near-equal loading

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."